GAREL'S UPSTAIRS MAPS ON THE DIFFERENTIAL FORMS OF A LOCALLY QUASI-FINITE SYNTOMIC MORPHISM, CONSTRUCTED ON THE NICE SUBCATEGORY AND PROPAGATED THROUGH THE UNIVERSAL FAMILIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. Prefilter history (duty patch of 2026-10-14 12:4x): this candidate was previously dropped by the lane's no-proof prefilter because its 10.6 kB proof contains the stray phrase 'follows easily' while merely remarking that a compatibility is easy; the proof was re-read by hand and judged complete (a five-stage structural argument, see the roadmap), so it is collected. The filter in next4.py now gates those phrase tests by length and sends any hit on a proof of 800 bytes or more to REVIEW instead of dropping it.

Allowed standard prerequisites (editorial).

- The analogous construction for the Tate map, whose proof this one follows. Discriminants, proposition discriminant-proposition-tate-map (invoked at proof line 3755). The author suggests reading that proof first, the present argument being very similar in shape.
- The category of locally quasi-finite syntomic morphisms, the base change lemmas for the maps, and the existence of compatible maps. Same chapter derham.tex, labels lemma-base-change-Garel-upstairs (invoked at proof lines 3775, 3818 and 3831) and lemma-existence-base-change-Garel-upstairs (invoked at proof line 3820). They formalise the statement as a rule on the category $\mathcal{C}$ compatible with base change morphisms, and provide the local existence of compatible maps used to glue and to transfer the construction.

Date: stacks-project:derham.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- The local description of a locally quasi-finite syntomic morphism and of the discriminant. Discriminants, label discriminant-lemma-syntomic-quasi-finite part (5) (invoked at proof lines 3790 and 3948), Remark \ref{remark-local-description} (invoked at proof line 3791), Section \ref{discriminant-section-tate-map} (invoked at proof line 3796) and Remark discriminant-remark-local-description-delta (invoked at proof line 3861). They give the local models on which the maps c^p are first constructed, and identify the etale locus with the nonvanishing locus of $\delta(\NL_{Y/X})$, which is what makes the maps unique after inverting δ.
- The universal locally quasi-finite syntomic families and the reduction to them. Discriminants, Example discriminant-example-universal-quasi-finite-syntomic (invoked at proof line 3852), label discriminant-lemma-universal-quasi-finite-syntomic-etale (invoked at proof line 3858) and label discriminant-lemma-locally-comes-from-universal (invoked at proof line 3870), and Morphisms, label morphisms-lemma-etale-at-point part (8) (invoked at proof line 3863). They show the universal families are objects of the nice subcategory (irreducible regular bases, nonvanishing δ) and that every object of $\mathcal{C}$ maps to one of them locally.
- Affine manipulations for the claim: standard opens, fibre products of rings and flatness. Properties, label properties-lemma-standard-open-two-affines (invoked at proof line 3910) and More on Algebra, labels more-algebra-lemma-module-over-fibre-product (invoked at proof line 3937) and more-algebra-lemma-properties-algebras-over-fibre-product (invoked at proof line 3945). They let the two competing base changes be compared in an affine chart, produce the auxiliary Y'/X' by a fibre product of rings and show it is again an object of $\mathcal{C}$.

Source reference. Original reference 0FK8: sources/stacks-chow/tags.txt, source lines 19453–19453. Navigation pointer for \ref{discriminant-example-universal-quasi-finite-syntomic} used inside the counted span (line 3852); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FKA: sources/stacks-chow/tags.txt, source lines 19455–19455. Navigation pointer for \ref{discriminant-lemma-locally-comes-from-universal} used inside the counted span (line 3870); the argument that consumes it is retained in the span.

Source reference. Original lemma 0BWE: sources/stacks-chow/tags.txt, source lines 14942–14942. Navigation pointer for \ref{discriminant-lemma-syntomic-quasi-finite} used inside the counted span (line 3790); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FK9: sources/stacks-chow/tags.txt, source lines 19454–19454. Navigation pointer for \ref{discriminant-lemma-universal-quasi-finite-syntomic-etale} used inside the counted span (line 3858); the argument that consumes it is retained in the span.

Source reference. Original proposition 0FKD: sources/stacks-chow/tags.txt, source lines 19458–19458. Navigation pointer for \ref{discriminant-proposition-tate-map} used inside the counted span (line 3755); the argument that consumes it is retained in the span.

Source reference. Original remark 0FKC: sources/stacks-chow/tags.txt, source lines 19457–19457. Navigation pointer for \ref{discriminant-remark-local-description-delta} used inside the counted span (line 3861); the argument that consumes it is retained in the span.

Source reference. Original section 0FKB: sources/stacks-chow/tags.txt, source lines 19456–19456. Navigation pointer for \ref{discriminant-section-tate-map} used inside the counted span (line 3796); the argument that consumes it is retained in the span.

Source reference. Original reference 0H9E: sources/stacks-derham/pinned-original-source.tex, source lines 3457–3457. Navigation pointer for \ref{item-base-change} used inside the counted span (line 3748); the argument that consumes it is retained in the span.

Source reference. Original reference 0H9C: sources/stacks-derham/pinned-original-source.tex, source lines 3448–3448. Navigation pointer for \ref{item-degree-zero} used inside the counted span (line 3747); the argument that consumes it is retained in the span.

Source reference. Original reference 0H9D: sources/stacks-derham/pinned-original-source.tex, source lines 3452–3452. Navigation pointer for \ref{item-multiplicative} used inside the counted span (line 3747); the argument that consumes it is retained in the span.

Source reference. Original lemma 0H9G: sources/stacks-derham/pinned-original-source.tex, source lines 3623–3623. Navigation pointer for \ref{lemma-base-change-Garel-upstairs} used inside the counted span (line 3775); the argument that consumes it is retained in the span.

Source reference. Original lemma 0H9H: sources/stacks-derham/pinned-original-source.tex, source lines 3687–3687. Navigation pointer for \ref{lemma-existence-base-change-Garel-upstairs} used inside the counted span (line 3820); the argument that consumes it is retained in the span.

Source reference. Original lemma 07RU: sources/stacks-chow/tags.txt, source lines 9879–9879. Navigation pointer for \ref{more-algebra-lemma-module-over-fibre-product} used inside the counted span (line 3937); the argument that consumes it is retained in the span.

Source reference. Original lemma 08KQ: sources/stacks-chow/tags.txt, source lines 10893–10893. Navigation pointer for \ref{more-algebra-lemma-properties-algebras-over-fibre-product} used inside the counted span (line 3945); the argument that consumes it is retained in the span.

Source reference. Original lemma 02GU: sources/stacks-chow/tags.txt, source lines 3241–3241. Navigation pointer for \ref{morphisms-lemma-etale-at-point} used inside the counted span (line 3863); the argument that consumes it is retained in the span.

Source reference. Original lemma 0H9B: sources/stacks-chow/tags.txt, source lines 21521–21521. Navigation pointer for \ref{properties-lemma-standard-open-two-affines} used inside the counted span (line 3910); the argument that consumes it is retained in the span.

Source reference. Original remark 0FL9: sources/stacks-derham/pinned-original-source.tex, source lines 3585–3585. Navigation pointer for \ref{remark-local-description} used inside the counted span (line 3791); the argument that consumes it is retained in the span.

Lemma 0.1. *There exists a unique rule that to every locally quasi-finite syntomic morphism of schemes $f : Y \to X$ assigns $\mathcal{O}_Y$-module maps*

$$c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

for $p \geq 0$ satisfying (0H9C), (0H9D), and (0H9E). In particular, the composition of $c^p_{Y/X}$ with $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ is multiplication by $\delta(NL_{Y/X})$.

Proof. This proof is very similar to the proof of Discriminants, Proposition 0FKD and we suggest the reader look at that proof first.

Let us reformulate the statement. Consider the category $\mathcal{C}$ whose objects, denoted Y/X, are locally quasi-finite syntomic morphism $f : Y \to X$ of schemes and whose morphisms $b/a : Y'/X' \to Y/X$ are commutative diagrams

$$\begin{array}{ccc} Y' & \xrightarrow{b} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{a} & X \end{array}$$

which induce an isomorphism of Y' with an open subscheme of $X' \times_X Y$. The lemma means that for every object Y/X of $\mathcal{C}$ we have maps $c^p_{Y/X}$, $p \geq 0$ with properties (0H9C) and (0H9D), and for every morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ we have that $c^p_{Y/X}$ and $c^p_{Y'/X'}$ are compatible as in Lemma 0H9G.

Given Y/X in $\mathcal{C}$ and $y \in Y$ we can find an affine open $V \subset Y$ and $U \subset X$ with $f(V) \subset U$ such that there exists some maps

$$\Omega^p_{Y/\mathbf{Z}}|_V \longrightarrow \left(f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})\right)|_V$$

with properties (0H9C) and (0H9D). This follows from picking affine opens as in Discriminants, Lemma 0BWE part (5) and then using the construction in Remark 0FL9.

Note that the étale locus of f is exactly the set of points where $\delta(NL_{Y/X})$ does not vanish, see discussion in Discriminants, Section 0FKB. In particular $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ becomes an isomorphism after inverting $\delta(NL_{Y/X})$. We conclude that if $\Omega^p_{X/\mathbf{Z}}$ is finite locally free and the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero, then the local maps constructed in the previous paragraph are unique and automatically glue!

Let $\mathcal{C}_{nice} \subset \mathcal{C}$ denote the full subcategory of Y/X such that

(1) $\Omega_{X/\mathbf{Z}}$ is locally free, and
(2) the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero.

By the remarks in the previous paragraph, we see that for any object Y/X of $\mathcal{C}_{nice}$ we have unique maps $c^p_{Y/X}$, $p \geq 0$ satisfying conditions (0H9C) and (0H9D). If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}_{nice}$, then the maps $c^p_{Y/X}$ and $c^p_{Y'/X'}$ are compatible as in Lemma 0H9G: namely, locally there do exist compatible maps by Lemma 0H9H and the uniqueness just mentioned shows these maps agree with $c^p_{Y/X}$ and $c^p_{Y'/X'}$. In other words, we have solved the problem on the full subcategory $\mathcal{C}_{nice}$. For Y/X in $\mathcal{C}_{nice}$ we continue to denote $c^p_{Y/X}$ the solution we've just found.

In fact, more generally, suppose we have a morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ such that Y/X is an object of $\mathcal{C}_{nice}$. The same argument, but this time using the uniqueness in Lemma 0H9G, tells us there are maps $c^p_{Y'/X'}$, $p \geq 0$ satisfying conditions (0H9C) and (0H9D) compatible with the already constructed maps $c^p_{Y/X}$. Let us call this the base change $(b/a)^*c^p_{Y/X}$.

Consider morphisms

$$Y_1/X_1 \xleftarrow{b_1/a_1} Y/X \xrightarrow{b_2/a_2} Y_2/X_2$$

in $\mathcal{C}$ such that Y_1/X_1 and Y_2/X_2 are objects of $\mathcal{C}_{nice}$. **Claim.** The two base changes $(b_i/a_i)^* c^p_{Y_i/X_i}$, $i = 1, 2$ are equal. We will first show that the claim implies the lemma and then we will prove the claim.

Let $d, n \geq 1$ and consider the locally quasi-finite syntomic morphism $Y_{n,d} \to X_{n,d}$ constructed in Discriminants, Example 0FK8. Then $Y_{n,d}$ and $Y_{n,d}$ are irreducible schemes of finite type and smooth over $\mathbf{Z}$. Namely, $X_{n,d}$ is a spectrum of a polynomial ring over $\mathbf{Z}$ and $Y_{n,d}$ is an open subscheme of such. The morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite syntomic and étale over a dense open, see Discriminants, Lemma 0FK9. Thus $\delta(NL_{Y_{n,d}/X_{n,d}})$ is nonzero: for example we have the local description of $\delta(NL_{Y/X})$ in Discriminants, Remark 0FKC and we have the local description of étale morphisms in Morphisms, Lemma 02GU part (8). Now a nonzero section of an invertible module over an irreducible regular scheme has vanishing annihilator. Thus $Y_{n,d}/X_{n,d}$ is an object of $\mathcal{C}_{nice}$.

Let Y/X be an arbitrary object of $\mathcal{C}$. Let $y \in Y$. By Discriminants, Lemma 0FKA we can find $n, d \geq 1$ and morphisms

$$Y/X \leftarrow V/U \xrightarrow{b/a} Y_{n,d}/X_{n,d}$$

of $\mathcal{C}$ such that $V \subset Y$ and $U \subset X$ are open. Then we have the base change $c^p_{V/U} = (b/a)^* c^p_{Y_{n,d}/X_{n,d}}$. The claim guarantees these locally constructed maps $c^p_{V/U}$ glue! Thus we get a well defined global maps $c^p_{Y/X}$ with properties (0H9C) and (0H9D). Finally, let $b/a : Y'/X' \to Y/X$ be an arbitray morphism of $\mathcal{C}$. We have the maps $c^p_{Y'/X'}$, $p \geq 0$ and $c^p_{Y/X}$, $p \geq 0$ constructed in this paragraph. To check they are compatible as in Lemma 0H9G we may work locally on Y'/X' and Y/X. Thus we may assume that there exists a morphism $Y/X \to Y_{n,d}/X_{n,d}$. By construction, both the family $c^p_{Y'/X'}$ and the family $c^p_{X/Y}$ are compatible with the map towards $Y_{n,d}/X_{n,d}$. It follows easily from this (and the uniqueness in Lemma 0H9G) that $c^p_{Y'/X'}$ is compatible with $c^p_{Y/X}$. Thus it remains to prove the claim.

In the rest of the proof we prove the claim. We may pick a point $y \in Y$ and prove the maps agree in an open neighbourhood of y. Thus we may replace Y_1, Y_2 by open neighbourhoods of the image of y in Y_1 and Y_2. Therefore we may assume Y, X, Y_1, X_1, Y_2, X_2 are affine, say they are the spectra of rings B, A, B_1, A_1, B_2, A_2. Picture

$$\begin{array}{ccccc} B_1 & \longrightarrow & B & \longleftarrow & B_2 \\ \uparrow & & \uparrow & & \uparrow \\ A_1 & \longrightarrow & A & \longleftarrow & A_2 \end{array}$$

By assumption the spectrum of B is an affine open of both the spectrum of $A \otimes_{A_1} B_1$ and $A \otimes_{A_2} B_2$. Shrinking more we may assume there exist elements $g_i \in A \otimes_{A_i} B_i$ such that our maps give isomorphisms $(A \otimes_{A_i} B_i)_{g_i} = B$, see Properties, Lemma 0H9B. Let x_α, y_β be a sufficiently large collection of variables such that we may choose surjections

$$A'_1 = A_1[x_\alpha] \to A \quad \text{and} \quad A'_2 = A_2[y_\beta] \to A$$

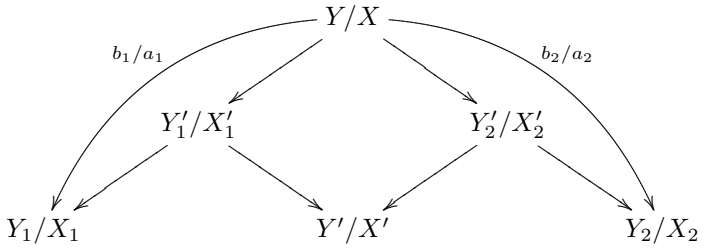
of A_1 and A_2-algebras. Then we can choose lifts $h_i \in A'_i \otimes_{A_i} B_i$ of $g_i \in A \otimes_{A_i} B_i$ and we consider the diagram

$$\begin{array}{ccccc} (A'_1 \otimes_{A_1} B_1)_{h_1} & \longrightarrow & B & \longleftarrow & (A'_2 \otimes_{A_1} B_2)_{h_2} \\ \uparrow & & \uparrow & & \uparrow \\ A'_1 & \longrightarrow & A & \longleftarrow & A'_2 \end{array}$$

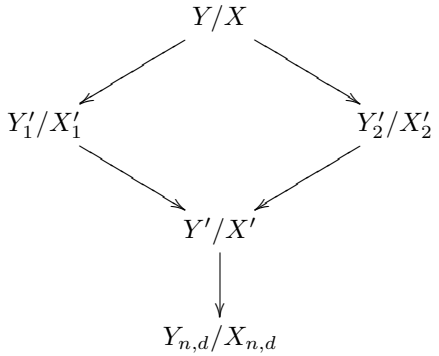
By construction the two squares are cocartesian. Next, we consider the ring map

$$A' = A'_1 \times_A A'_2 \longrightarrow B' = (A'_1 \otimes_{A_1} B_1)_{h_1} \times_B (A'_2 \otimes_{A_1} B_2)_{h_2}$$

By More on Algebra, Lemma 07RU we have $A'_1 \otimes_{A'} B' = (A'_1 \otimes_{A_1} B_1)_{h_1}$ and $A'_2 \otimes_{A'} B' = (A'_2 \otimes_{A_2} B_2)_{h_2}$. In particular the fibres of the morphism $Y' = \mathrm{Spec}(B') \to \mathrm{Spec}(A') = X'$ are open subschemes of base changes of the fibres of the maps $Y_i \to X_i$, hence finite and local complete intersections (see Discriminants, Lemma 0BWE). By More on Algebra, Lemma 08KQ the ring map $A' \to B'$ is flat and of finite presentation. Thus by Discriminants, Lemma 0BWE part (3) we conclude that Y'/X' is an object of $\mathcal{C}$. Consider now the commutative diagram



where Y'_i/X'_i corresponds to $A'_i \to (A'_i \otimes_{A_i} B_i)_{h_i}$. Note that Y'_i/X'_i is an object of $\mathcal{C}_{nice}$ as it obtained by taking an open subscheme of the product of an infinite dimensional affine space with Y_i/X_i; small detail omitted. In particular, the pullback of $c^p_{Y_i/X_i}$ via (b_i/a_i) is the same as the pullback of $c^p_{Y'_i/X'_i}$ via $Y/X \to Y'_i/X'_i$. Now we would be done if Y'/X' is an object of $\mathcal{C}_{nice}$, but this is almost never the case. Namely, then pulling back $c^p_{Y'/X'}$ around the two sides of the square, we would obtain the desired conclusion. To get around the problem that Y'/X' is not in $\mathcal{C}_{nice}$ we note the arguments above show that, after possibly shrinking all of the schemes $X, Y, X'_1, Y'_1, X'_2, Y'_2, X', Y'$ we can find some $n, d \geq 1$, and extend the diagram like so:



and then we can use the already given argument by pulling back from $c^p_{Y_{n,d}/X_{n,d}}$. This finishes the proof. □